

IAEA proficiency tests for determination of radionuclides in sea water



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A B S T R A C T

The International Atomic Energy Agency organised four proficiency tests between 2012 and 2015 to test the performance of participating laboratories in an analysis of radionuclides in sea water samples. These exercises were initiated to support IAEA Member States in sea water analyses of tritium, strontium-90 and caesium isotopes in relation to the accident at the Fukushima Daiichi nuclear power station, in March 2011, and subsequent contamination of the marine environment.

1. Introduction

In total four annual IAEA proficiency tests were organised between 2012 and 2015 to test the performance of participating laboratories in an analysis of radionuclides in sea water samples. These exercises were initiated to support IAEA Member States in sea water analyses of tritium, strontium-90 and caesium isotopes in relation to the accident at the Fukushima Daiichi nuclear power station, in March 2011, and subsequent contamination of the marine environment. The first exercise was organised in 2012 as part of the IAEA Technical Cooperation project RAS/7/021 entitled Marine Benchmark Study on the Possible Impact of the Fukushima Radioactive Releases in the Asia-Pacific Region and was followed by three exercises in 2013–2015. Two other exercises were organised in 2014 and 2015 in cooperation with the Nuclear Regulation Authority (NRA) in Japan. A total of 59 invited laboratories from 35 countries participated in these six exercises, including 13 laboratories from Japan. In addition, several laboratories taking part in the IAEA Technical Cooperation project RAS/7/021 or from OSPAR and HELCOM contracting parties participated in these exercises (Table 1).

2. Method

In the first exercise (RML-2012-01), each participant received 10 g of a 1 M HCl solution containing ^{134}Cs and ^{137}Cs (with massic activities only known to the IAEA) and a 450 ppm Cs carrier. The ampoules were intended to be diluted with exactly 100 L of local sea water, resulting in massic activities in the order of 0.07 Bq kg⁻¹ and 0.10 Bq kg⁻¹, respectively.

For all subsequent exercises, samples containing 5 L of filtered and acidified Mediterranean Sea water spiked by the IAEA with radionuclides were distributed to the participants, with the massic activities only known to the IAEA. The massic activities were traceable to a standard provided by a national standards laboratory. The combined massic activities in the exercise samples were lower than the natural activity level of ^{40}K in sea water (which is approximately 12 Bq kg⁻¹). The ^3H massic activities for the exercise samples were in the order of 3 Bq kg⁻¹, while the ^{90}Sr , ^{134}Cs and ^{137}Cs massic activities were between 0.1 Bq kg⁻¹ and 0.4 Bq kg⁻¹. These massic activities represent the magnitude for some of the activity levels reported for sea water collected within a 2 km radius from Fukushima Daiichi nuclear power station between 2012 and 2016.

The participants were asked to carry out an independent analysis of the sea water sample and to report the results of the massic activities together with their uncertainty at $k = 1$ (both expressed as Bq kg⁻¹) for a given reference time.

The evaluation methodology adopted for these exercises was slightly updated from the existing IAEA methodology (Osvath et al., 2016). The scoring system took into account the accuracy, precision and trueness of the reported data and included in the evaluation both the combined standard uncertainty of the IAEA value and the combined standard uncertainty reported by the participating laboratories. A result had to pass three statistical tests to be assigned the status “Accepted”, otherwise it was assigned the status “Warning” or “Not Accepted”.

The first step in the evaluation was the estimation of the bias. The relative bias between laboratory value x_{lab} and the assigned IAEA value x_{IAEA} was calculated as follows and is expressed as a percentage:

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Table 1
Overview the proficiency test exercises.

Exercise	Organised for	Participants	Radionuclides
IAEA-RML-2012-01 (IAEA, 2015a)	RAS/7/021	23	^{134}Cs and ^{137}Cs
IAEA-RML-2013-01 (IAEA, 2015b)	RAS/7/021	31	^{90}Sr , ^{134}Cs and ^{137}Cs
IAEA-RML-2014-01 (IAEA, 2015c)	NRA Japan	30	^3H , ^{90}Sr , ^{134}Cs and ^{137}Cs
IAEA-RML-2014-02 (IAEA, 2015d)	RAS/7/021	14	^{137}Cs
IAEA-RML-2015-01	NRA Japan	26	^3H , ^{90}Sr , ^{134}Cs and ^{137}Cs
IAEA-RML-2015-02	RAS/7/021	17	^{137}Cs

$$Bias_{\text{relative}} = \frac{x_{\text{lab}} - x_{\text{IAEA}}}{x_{\text{IAEA}}} \times 100\% \quad (1)$$

The absolute value of the relative bias was compared to the Maximal Accepted Relative Bias (MARB). Participants' results were scored as "Pass" for accuracy when:

$$|Bias_{\text{relative}}| \leq \text{MARB} \quad (2)$$

The MARB values are set by the IAEA before the exercise and their values are generally between 15% and 25%, depending on the perceived difficulty of the analysis. In these exercises, the MARB values for tritium and ^{90}Sr were set at 25%, while for ^{134}Cs and ^{137}Cs they were set at 20%. The precision P for each result was calculated according to the following equation:

$$P = \sqrt{\left(\frac{u_{\text{IAEA}}}{x_{\text{IAEA}}}\right)^2 + \left(\frac{u_{\text{lab}}}{x_{\text{lab}}}\right)^2} \times 100\% \quad (3)$$

The precision P was compared to the Limit of Accepted Precision (LAP). The participants' results were scored as "Pass" for precision when:

$$P \leq \text{LAP} \quad (4)$$

The LAP values are also set by the IAEA before the exercise and their values are generally between 15% and 25%, depending (i) on the perceived difficulty of the analysis and (ii) the magnitude of relative uncertainty of the IAEA value. In these exercises, the LAP values for tritium and ^{90}Sr were set at 25%, while for ^{134}Cs and ^{137}Cs they were set at 20%. The participants' results for trueness were scored as "Pass" when:

$$|Bias_{\text{relative}}| \leq \frac{x_{\text{lab}}}{x_{\text{IAEA}}} 2.58 \cdot P \quad (5)$$

where the factor of 2.58 is the critical value at a 99% confidence level. For the final evaluation, all three scores were combined (see Table 2).

The result was considered as "Accepted" if it passed all three tests. If the accuracy test was failed, the result was considered as "Not Accepted". If the accuracy test was passed but one of the other two tests was failed, the result was assigned the "Warning" status. The "Warning" status will reflect cases in which the reported result was close enough to the assigned property value (i.e. the absolute value of the relative bias was less than the MARB), but its associated uncertainty was deemed to be too large (e.g. the combination in quadrature of the

Table 2
Performance evaluation criteria.

Accuracy	Precision	Trueness	Final evaluation
Pass	Pass	Pass	Accepted
Pass	Fail	Pass	Warning
Pass	Pass	Fail	Warning
Fail	Pass/Fail	Pass/Fail	Not Accepted

Table 3
Overview of the evaluation results for each radionuclide in each individual exercise.

Exercise	Evaluation	^3H	^{90}Sr	^{134}Cs	^{137}Cs
RML-2012-01	Accepted			13 (57%)	14 (61%)
	Warning			4 (17%)	4 (17%)
	Not Accepted			6 (26%)	5 (22%)
RML-2013-01	Accepted		8 (61%)	21 (68%)	22 (74%)
	Warning		1 (8%)	3 (10%)	4 (13%)
	Not Accepted		4 (31%)	7 (22%)	4 (13%)
RML-2014-01 and RML-2014-02	Accepted	14 (67%)	16 (66%)	25 (63%)	27 (64%)
	Warning	1 (5%)	4 (17%)	6 (15%)	6 (14%)
	Not Accepted	6 (28%)	4 (17%)	9 (22%)	9 (22%)
RML-2015-01 and RML-2015-02	Accepted	15 (65%)	18 (78%)	26 (60%)	32 (76%)
	Warning	2 (9%)	2 (9%)	5 (12%)	3 (7%)
	Not Accepted	6 (26%)	3 (13%)	12 (28%)	7 (17%)

lab's relative uncertainty and the IAEA relative uncertainty exceeds the LAP) or either too small resulting in the relative bias being significantly different from zero (or, in other words, the reported result is significantly different from the assigned IAEA value).

3. Results

The results of the four annual exercises are presented in Table 3. The evaluation of the individual exercise results showed reasonably consistent results. For tritium (included in 2 exercises) and ^{90}Sr (included in 3 exercises) about two-thirds of the results were accepted, although not all participants reported these two radionuclides. A relatively small number of tritium and ^{90}Sr results were evaluated as "Warning", which reflect cases in which the reported laboratory results were close enough to the assigned IAEA values, but its associated uncertainty was deemed to be either too small or too large. For ^{134}Cs and ^{137}Cs , which were reported by practically all exercise participants, the range of accepted results was 57–68% and 61–76%, respectively. In some exercises, for ^{134}Cs significant negative biases of the combined participants' results were observed, while, in general, there were no significant biases for tritium, ^{90}Sr and ^{137}Cs . These observations are supported when the nuclide-specific results for all exercises are combined into single datasets (see Figs. 1–4). In these Figures, the relative biases are represented by the black points. The error bars represent the standard uncertainties of the bias (at $k = 1$), which is calculated as the product of the precision P and the ratio between x_{lab} and x_{IAEA} . The dotted lines represent the MARB values (any bias either below the lower MARB line or above the higher MARB line has the "Not Accepted" status).

Calculation of median values for the bias for each radionuclide resulted in a significant negative value at a 99% confidence level for ^{134}Cs , i.e. $-3.0(10)\%$, while the median values for bias for tritium, ^{90}Sr

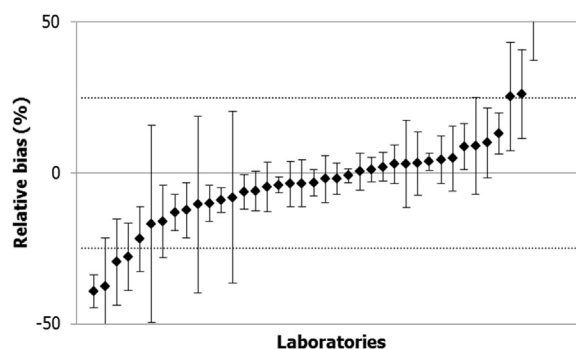


Fig. 1. ^3H bias (%) for the combined exercises. The error bars represent the standard uncertainties of the relative bias (with a coverage factor of $k = 1$). The dotted lines represent relative biases of $\pm 25\%$ (MARB).

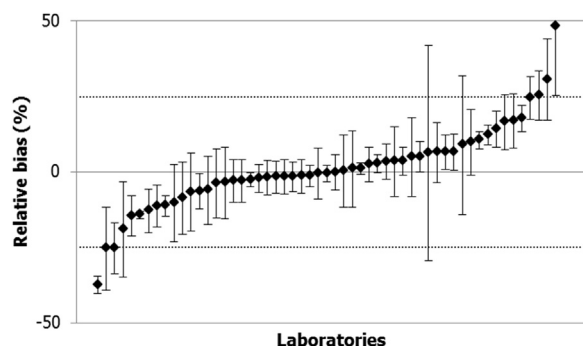


Fig. 2. ^{90}Sr bias (%) for the combined exercises. The error bars represent the standard uncertainties of the relative bias (with a coverage factor of $k=1$). The dotted lines represent relative biases of -25% and 25% (MARB).

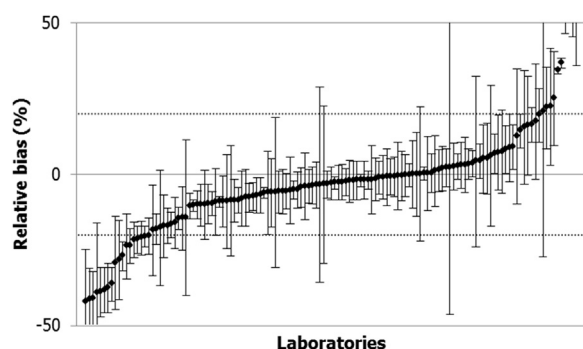


Fig. 3. ^{134}Cs bias (%) for the combined exercises. The error bars represent the standard uncertainties of the relative bias (with a coverage factor of $k=1$). The dotted lines represent relative biases of -20% and 20% (MARB).

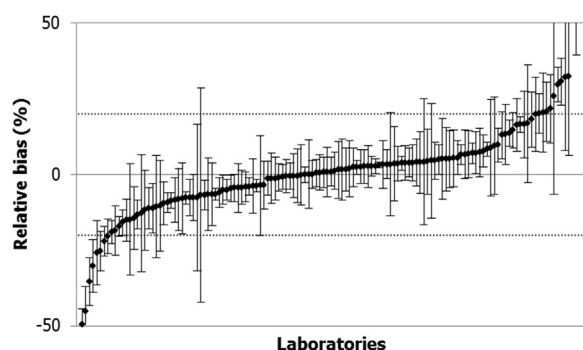


Fig. 4. ^{137}Cs bias (%) for the combined exercises. The error bars represent the standard uncertainties of the relative bias (with a coverage factor of $k=1$). The dotted lines represent relative biases of -20% and 20% (MARB).

and ^{137}Cs were not found to be significantly different from zero [$-2.0(23)\%$, $-0.4(17)\%$ and $1.0(10)\%$, respectively], where the numerical result is stated in the format $xx(yy)\%$, with the number in parentheses representing the numerical value of the uncertainty referring to the corresponding last digits of the quoted value. Coincidence summing is a problem for ^{134}Cs as it leads to signal loss and hence underestimation of the activity levels for this radionuclide. It is clear from the results that some participants did not make a sufficient correction for coincidence summing with 67% of the reported ^{134}Cs results showing a negative bias.

The subset of 13 Japanese participants that participated in IAEA-RML-2014-01 and IAEA-RML-2015-01 and that were nominated by the Nuclear Regulation Authority (NRA) in Japan, obtained very good results in both exercises. For tritium and ^{90}Sr , 100% and 91% of the results were accepted, while this was the case for 75% and 85% of the ^{134}Cs and ^{137}Cs results. The overall score for the Japanese results was 86% “Accepted”, 9% “Warning” and 5% “Not Accepted”. The median

values for bias for tritium, ^{90}Sr , ^{134}Cs and ^{137}Cs were not found to be significantly different from zero [$-1.4(24)\%$, $-1(4)\%$, $-2.2(25)\%$ and $4(3)\%$, respectively].

For ^3H , practically all participants performed a distillation of the sea water samples, followed by liquid scintillation counting (LSC). Some performed an electrolytic enrichment of tritium after the distillation (and obtained excellent results with small relative uncertainties between 1.1% and 4.8%). The direct measurements after distillation yielded larger relative uncertainties between 5% and 15% (see Fig. 1). Three participants reported relatively large relative uncertainties between 26% and 39% that failed the pre-set LAP value of 25% resulting in a “Warning” status.

For ^{90}Sr (Vadja and Kim, 2010), most participants used gas-flow proportional counting (GPC) of chemically separated ^{90}Y as the analysis technique, with a minority that used LSC/Cherenkov counting. Chemical separation techniques used included precipitations of Sr-oxalate or Sr-carbonate, barium chromate precipitations, nitric acid precipitations, cation-exchange chromatography, Sr-extraction chromatography and direct liquid-liquid extraction of ^{90}Y with HDEHP. No significant difference between the performances of the chemical separation techniques is apparent. The reported relative uncertainties (at $k=1$) were typically between 3% and 10% (see Fig. 2). One participant reported a relatively large relative uncertainty of 34% that failed the pre-set LAP value of 25% resulting in a “Warning” status. A few participants reported rather small relative uncertainties in the order of 1.2% to 3.5% resulting in their relative bias being significantly different from zero (resulting in a different “Warning” status).

For ^{134}Cs and ^{137}Cs (Aoyama et al., 2016), a majority of the participants (64%) measured the sea water samples directly with gamma spectrometry, while a minority of laboratories either used a pre-concentration technique (i.e. adsorption on ammonium molybdophosphate or hexacyanoferrate containing either copper, nickel or cobalt) to separate the caesium radionuclides from the sea water matrix followed by gamma spectrometry or they used a combination of the two techniques. No significant difference in the performance between the two techniques is apparent, although in general the reported relative uncertainties (at $k=1$) for the direct measurements were somewhat larger, typically between 5% and 15% for the direct measurements and between 2% and 8% for the pre-concentration methods. As can be seen from Figs. 3 and 4, a few participants reported relatively large relative uncertainties (in most cases for ^{134}Cs) between 22% and 48% that failed the pre-set LAP value of 20% resulting in (at least) a “Warning” status. A few participants reported rather small relative uncertainties as low as 0.6% resulting in their relative bias being significantly different from zero (resulting in a different “Warning” status).

4. Conclusion

The results of the IAEA sea water proficiency test exercises organised between 2012 and 2016 are satisfactory. The percentage of accepted results (approximately 70% for all participants) is more or less at the same level as for other environmental schemes such as the IAEA ALMERA exercises (Osvath et al., 2016) and the exercises organised by the National Physical Laboratory (NPL) (Harms and Gilligan, 2011). The ^{134}Cs results confirm the observations in these schemes that some participants do not make a sufficient correction for coincidence summing and hence tend to somewhat underestimate the ^{134}Cs activity levels. Another observation is the relative large range of reported uncertainties. Approximately 10% of results was evaluated as “Warning” as their reported uncertainty was deemed to be either too small or too large.

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